The Law of Graphic Equalisation — Physics Grounding

Change, c, the Reconciliation of Special and General Relativity, the Recovery of the Born Rule, the Dimensional Constants, and the Second Law

Matthew Parslow
Independent Researcher

Draft — draft-2026-08-27
27 August 2026

DRAFT — draft-2026-08-27 (27 August 2026). This document is under active revision and is circulated for comment, not as a finished result. It is a companion to *The Law of Graphic Equalisation — Theory* (referred to below as Graphic Equalisation), and inherits that paper's definitions, epistemic boundary, and level-of-claims. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This companion to Graphic Equalisation develops the physics groundings of the change-substrate. Its central move is that a frame's time is an inferred count of *cycles* and that the invariant speed c is the frame's own change-throughput unit, 1 by construction rather than by postulate; from that one economy the relativities, the dimensional constants, and the second law follow as *shapes*. Special relativity's c-invariance becomes a theorem; the two relativities reconcile as one fixed change-budget spent two ways, on motion or on change-load; the dimensional constants $(c, \hbar, k_{\mathrm{B}}, G)$ are each a frame's own base unit and their magnitudes are cross-frame artefacts; the uncertainty principle is the finite-aperture floor, the Landauer bound the information–energy conversion, quantization the aperture cell, and the Planck scale the frontier. The second law is no-copy accumulated, with entropy frame-relative, the demon exorcised by cost, and local order affordable as paid minting. On this one substrate quantum theory and relativity reconcile at the level of shape: c-invariance and the special–general reconciliation are theorems given the substrate's own units, and the Born rule is recovered — not forced — on the same primitives (closure, no-privilege, free continuation). The same primitives give entanglement as a superframe dimension, the double slit as possibility structure under superframe forcing, and identical particles as functional identity, while the seed, the gauge group, the dimensionless constants and the number of spatial dimensions are contingent rule-content answered by self-location. The claims are information-level, physics taken as an embodiment: the paper mechanises shapes and derives no field equation, no dimensionless constant, and no theory of quantum gravity.

1 Scope

This paper is a grounding, not a physical theory. It inherits Graphic Equalisation's position that the claims are about information — distinction, frame, admissibility, realisation — of which physical theory is one embodiment, and that a physical result bears on an information-level claim when it settles an informational fact. Everything below is at that level. The strong claims are structural: that there is a c, a $\hbar$, an uncertainty principle, a Landauer floor, a second law, and that special and general relativity reconcile as one change-economy. To these the quantum arc adds collapse as frame-relative realisation, entanglement as a superframe dimension, and — the one place a number is reached — the Born rule, *recovered* given contingent rule-content rather

than derived as a necessity, with the uniqueness supplied by cited theorems. The weak part is otherwise anything carrying a number: no field equations, no quantitative metric, no exact Lorentz factor, no dimensionless constant, and no theory of quantum gravity. Each grounding is a *promoted* constraint exposed to falsification, in the manner of Graphic Equalisation's collapse bridge, not a result asserted from within.

The minimal-universe argument that licenses the contingency verdicts below — a rock is a universe, and the minimal universe does without time, the gauge group, the constants and the dimensions, so none can be compulsory — together with the run/record axis, free will, personal identity, and death, is developed in the companion paper *The Law of Graphic Equalisation — Mind and Self Grounding*, which treats philosophy of mind as applied physics of the run.

2 c as the change-to-time rate

Graphic Equalisation establishes that time is not a substrate dimension but is *inferred* by a frame, whose unit is a *cycle* — one traversal of a closed rule-loop, of no fixed length. Once time is the inferred count of cycles, the invariant speed c has a natural reading, consistent with the missing-denominator result (there is no rate *in* substrate-time, since there is no substrate-time to divide by). c is the fixed *rate* at which change is realised — per unit of the substrate's throughput, a cycle consuming a variable amount of it — from which a frame's time is inferred. So c is prior to time: the conversion between change-count and inferred duration, which is exactly what a constant relating the temporal and spatial parts of an interval is.

Proposition 2.1 (c is each frame's own unit, hence invariant by construction)**.** *c is each frame's own base rate, its change-throughput, so every frame measures its own c as the unit,* 1 *— and a unit cannot be measured as anything else. Its invariance is therefore a theorem, not a postulate: not that all frames happen to agree on a speed, but that each is reporting its own unit.*

A value *other* than 1 is a cross-frame quantity: the familiar number appears only when one frame is compared with another — change-throughput against a spatial metric, seconds against metres — which is where the continuous lives, in the maps between frames rather than in a frame's own discrete states. One can give time a number at all only by comparing frames; in its own frame c is the unit, and its particular value (299 792 458 in SI) is an artefact of the comparison, not a fact about the substrate. And c is a *limit* because a signal is a sequence of changes and none can be realised faster than the frame's throughput.

It matters what is and is not being stipulated here, because the nearest neighbouring account stipulates precisely what this one does not. Discrete-substrate derivations of relativity typically assign a metric quantity and obtain c-constancy from the assignment: in the Wolfram model, constancy of the speed of light is, in the author's own words, "enforced axiomatically by our definition of the edge lengths in causal graphs" (Gorard, 2020a). Nothing is assigned here. A cycle is a closure event, not a length: how much substrate throughput one traversal consumes depends jointly on the rules and on the graph those rules are applying to, so there is no fixed unit available to be set. A frame reads its own c as 1 because it counts its own cycles and possesses no in-frame length against which to compare them — the unit is unavailable, not chosen.

The difference is not verbal, because it is where the empirical content sits. A stipulated edge length is constant by construction and can therefore predict nothing further. A cycle whose length *varies* with rules and graph predicts a great deal: that concentrating change-load lengthens cycles and slows clocks, that mass is throughput reserved for internal change, and that the whole of Section 3 follows as consequences rather than as further postulates. The honest residual is that this does not make c-invariance an empirical prediction — a frame counting its own cycles will read 1 whatever the substrate does — and the paper does not claim it as one. What it claims is that the invariance costs nothing to assume because it is a

fact about self-measurement, and that the variation in cycle length, which is not a fact about self-measurement, carries the physics.

3 The change-budget: speed, dilation, mass, aperture

The fixed throughput c splits between *spatial motion* and *internal change* (proper time), and a speed is the fraction spent on motion. At rest the whole budget is internal change and the clock ticks fully; approaching c moves budget to motion, leaving less for internal change, so the clock slows — time dilation as budget re-allocation — and at c the whole budget is motion and no proper time remains. Mass is the budget *reserved* for internal change, so a massive graph can never put all of it into motion: it asymptotes to c and the energy to approach diverges, while a massless one reserves none and *is* at c, with no proper time.

Because observation is funded by internal-change throughput, approaching c coarsens a frame's *aperture*. The transformation between frames in relative motion — time dilation and length contraction — is an aperture transformation between different budget-splits, and each frame's aperture self-calibrates so that its own c still reads 1. The Lorentz factor is the aperture change that holds $c = 1$ across the split. (The constant-magnitude four-velocity of Minkowski geometry (Minkowski, 1909) is this budget-split stated as a metric fact rather than as a mechanism.)

The same c bounds *concurrency*, which gives the high-energy regime a reading. Within a frame, change is self-change and strictly ordered, so forcing two changes onto one rule-step is the paid exception; a particle accelerator is the machine that pays it, spending energy to push particles' interactions to the same step, where the frame — built to sequence — cannot order them and the interaction is *decoherent* in that frame. Two changes cannot be crowded closer than a step, and forcing them there costs more the closer one gets: the diverging energy above is this same ceiling read at the change-rate. The strong reading is "energy is the price of forced concurrency, and forced concurrency is frame decoherence"; anything with a number is weak.

4 Special and general relativity reconciled

The two relativities are one mechanism, because the one fixed budget c can be drained two ways.

Proposition 4.1 (SR and GR are one change-budget spent two ways)**.** *Special relativity is the budget spent on* motion*: a moving graph puts throughput into spatial motion, leaving less for internal change, so its clock slows. General relativity is the budget spent on* change-load*: mass is throughput tied up in internal change, and concentrating change-load (dense mass) consumes the local budget, so local clocks slow. Both dilations are the same act — the one fixed budget drained, by motion or by mass, each leaving less for proper time.*

The equivalence principle is the seam. Accelerating (changing one's motion) and standing in a gravitational field (change-load) draw on the budget identically, so inertial mass (resistance to a change of motion — the impulse relation of the Narrative-momentum paper, with mass as the exchange rate) and gravitational mass (change-load consuming throughput and slowing local time) are one quantity, "how much change this matter loads onto the ruleset," measured two ways. General relativity assumes their equality and does not explain it; here they coincide by construction.

Corollary 4.2 (Relativity reconciles because c is defined, not postulated)**.** *Because c is here* defined *(Proposition 2.1) rather than postulated, special relativity's postulate of c-invariance becomes a theorem, and its effects follow as consequences rather than as separate surprises, with the classical paradoxes dissolving into aperture differences — each moving clock right, each reading its own $c = 1$. Special and general relativity then reconcile (Proposition 4.1) as velocity- and gravity-dilation of one budget, joined at the equivalence principle.*

5 Relativity and collapse

Relativity also reconciles with the collapse picture of quantum measurement, by a division Graphic Equalisation already carries (Section 2): the continuous in the maps between frames, the discrete in a frame's own states. The continuous geometry of general relativity is a cross-frame *map*; the collapse of quantum measurement is the in-frame *event*; and c — the change-to-time rate — is the seam joining the discrete event to the continuous metric. The two are the map and the event of one change-substrate, not rival ontologies, and the standing incompatibility is the artefact of treating them as rivals. This is a claim about the *shape* of the reconciliation, not a derivation: that the field equations follow is not asserted, and no theory of quantum gravity is offered.

6 The measurement problem is frame-relative collapse

Collapse in Graphic Equalisation is a local write that realises one continuation *relative to the frame that interacts*. It is not a physical process needing a mechanism — no dynamical collapse, no spontaneous localisation — but the in-frame realisation that occurs when frames couple. So there is no frame-independent "when": each frame realises its own outcome upon interaction, and a frame that has not interacted holds no collapse. This is Axiom -1 and relational quantum mechanics (Rovelli, 1996) — facts indexed to interactions, not assigned from nowhere.

Proposition 6.1 (Wigner's friend is the map and the event of two frames)**.** *The friend measures the system and realises an outcome — collapse relative to the friend, who interacted. Wigner, not having interacted with friend-and-system, holds relative to himself the continuous unitary* map *(Section 5). Both are correct: the friend holds the in-frame* event*, Wigner the cross-frame* map*. The apparent contradiction requires a single frame-independent fact about whether collapse occurred, which Axiom* -1 *forbids; without that demand the friend's definite result and Wigner's superposition coexist without strain.*

Corollary 6.2 (Any coupling is a measurement; outcomes are definite; no special observer)**.** *Any coupling is a measurement relative to the coupled frames, so no conscious or privileged observer is required and none is distinguished — a plate, a rock, or a friend measure relative to whatever they couple to. Outcomes are definite because collapse realises* one *continuation, and the basis in which they are definite is set by the distinctions the interaction makes — the aperture of the coupling, which is decoherence selecting the pointer basis. There is no cut and no collapse mechanism to locate. A superframe* applying *a change to a subframe is not itself such a coupling: it makes no distinction the subframe holds, so it realises nothing for that subframe — which is why the superframe forcing of Section 9 leaves an interference pattern intact while a which-path coupling destroys it.*

This dissolves the measurement *problem* — the when, the how, the who, and the definiteness. That outcomes are realised frame-relatively is the shape; how their probabilities are weighted is a separate question, taken up in Section 8.

7 Entanglement is dimensional access through frames

An entangled pair is one system in the superframe, described by a joint dimension — the correlation — that is a first-class quantity of the superframe and NULL in each subframe individually (Graphic Equalisation's higher-frame first-class structure). Each subframe has dimensional access to it only through its own boundary, and only in part; the full dimension lives one level up, along the structural-depth axis rather than the spatial one.

Proposition 7.1 (A subframe's local state is mixed because the purity is a superframe dimension)**.** *Described alone, an entangled particle has no pure state — it is mixed — because its pure description requires the joint dimension, which is* NULL *in its subframe. Tracing out the partner is not discarding locally-available information; the missing information was never a subframe dimension. The reduced state is mixed exactly because the purity is a superframe dimension, inaccessible from below.*

Corollary 7.2 (Nonlocality is up, not across; Bell forbids only subframe-local variables)**.** *The connection between the parties is not* across *the space between them but* up*, to the superframe they are both subframes of. No cross-frame write passes between the subframes; each accesses the same superframe dimension locally, and the joint fact appears only when their accesses are compared (a further coupling), so no signal is sent. Bell forbids* local *— subframe-level — hidden variables (Bell, 1964), and the framework has none: the correlating variable is a superframe dimension, so the violation is expected, not paradoxical. Kochen–Specker contextuality (Kochen and Specker, 1967) is the access itself — measuring projects the superframe dimension into the subframe's aperture, so a realised value is context-dependent by construction.*

Remark (Spookiness is an aperture artefact: the light-switch identity)*.* The confusion entanglement provokes is identical in structure to that of a person who has never seen a light switch and does not understand electricity: flip a switch, a far light obeys, with no connection inside the aperture, so action at a distance is inferred. In both cases the inference is wrong for the same reason — there is a connection, through a dimension the observer cannot see: the wiring in the walls for the switch (the opaque runner), the superframe's joint dimension for entanglement (structural depth). Spookiness is therefore a property of the aperture, not of the phenomenon; a switch is mundane once one sees the wiring, entanglement once one sees the superframe. This is the sequence science has run before — magnetism was action at a distance until the field, gravity until curvature, disease until germs — each "spooky" a real connection through a not-yet-visible dimension, each dissolved by the aperture widening rather than the phenomenon changing. "We do not understand entanglement" is structurally "I do not understand the light switch."

Honest boundary: seeing the connecting dimension dissolves the *spookiness*, not the *numbers* — the Tsirelson bound, the correlation strength and the Born weights are not derived; they become ordinary physics questions rather than spooky ones.

8 Amplitudes are a rule on Narrative; recovering the Born rule

An amplitude is not a foreign number-structure but the directional Narrative of the Narrative-momentum paper, acted on by a rule. That paper's directional momentum $p = \phi_W(\mathcal{N})\,\hat{\mathbf{d}}$ is a magnitude $\phi(\mathcal{N})$ — the operative-history measure — times a direction $\hat{\mathbf{d}}$. The complex amplitude is therefore the directional Narrative, its dynamics a rule applied to it, and its phase the direction $\hat{\mathbf{d}}$ read as position in the cycle (Section 2). The complex structure of quantum amplitudes is thus grounded, not imported.

Proposition 8.1 (Closure makes the weight a conjugate pair)**.** *A realised outcome is a* closed *cycle — the realised "now" is one traversal of the closure loop, and a loop that closes. A probability is the weight of a realised outcome, hence of a closed loop: the Narrative carried forward and carried back to close it, the amplitude and its return. The weight is then the amplitude applied twice, amplitude times its conjugate, $|amplitude|^2$. The Born square would on this reading be the closure itself — a realisation must close its loop, and closing requires the directional Narrative both ways.*

This is a candidate, not a derivation, and a real advance on leaving the amplitude a blank.

An attempt to close it

The candidate can be pushed further than a candidate, though not all the way. Take the amplitude as a directed magnitude — a Narrative's size r times a phase — and follow four steps, marking what each earns. *This subsection is an attempt, flagged as such, not a claimed result.*

Complex (motivated). The phase is position on the realisation cycle, and a cycle's traversal parameter is one-dimensional: $S^1 = U(1)$. So an amplitude is $r\,e^{i\theta}$, a complex number — real would be a phase-less (cycle-less) magnitude, and the one-dimensional closure loop gives a single angle, not the three of a quaternionic phase. Motivated by closure, not forced.

Linear (a candidate mechanism, not a proof). Independent contributions to one outcome superpose by addition. This has a candidate mechanism rather than a bare posit. Free evolution is the amplitude's own Narrative, and a Narrative is what a subgraph runs on an *empty* rule set — the free continuation carried forward when no rule fires (the momentum of Narrative-momentum, the free-continuation default of Section 12). With no rule present, nothing *couples* the components of a superposition; each free-continues independently, and uncoupled independent continuation *is* linear superposition, since a nonlinearity would need a rule reading one component and acting on another — exactly what the empty set lacks. Linearity is on this reading the signature of the empty rule set. What stays a residual is the rigorous step from "uncoupled" to full additivity and norm-preservation (unitarity), and the linearity of the imposed supergraph map is not handed over for free: the mechanism motivates linearity, it does not yet prove it.

Phase-independent (from no privilege). A cyclic direction has no privileged origin, so a global phase rotation is a change of nothing (Graphic Equalisation's no-privilege principle applied to the phase). The realised probability must therefore be invariant under a global $U(1)$, hence a function of the modulus alone, $P = f(|a|)$. This step is earned, and it is the no-privilege thesis of the mind companion doing physics work.

Squared (forced, given the above). Two alternatives with no privileged relative phase — decohered, their relative phase averaged over — are classically exclusive, so their probabilities must add. Averaging the combined modulus over the relative phase,

$$\left\langle f(|a\,e^{i\theta} + b|)\right\rangle_\theta = f(|a|) + f(|b|),$$

since the cross term $2\,\mathrm{Re}(a^* b\,e^{i\theta})$ averages to zero. The square, $f(x) \propto x^2$, satisfies it, and the obvious alternatives do not: a quartic overshoots by the surviving $\langle\cos^2\rangle = \frac{1}{2}$ term, the linear undershoots ($a = b = 1$ gives $4/\pi \neq 2$), the square lands exactly; normalisation fixes the constant. This selects the square among the natural candidates rather than proving it unique — the uniqueness is what Gleason's theorem supplies below, over all measurement contexts. The weight is real because closure (a returned cycle) is net-zero phase.

The Gleason residual, reconsidered. The internal Narrative is relative: it is the vector of the frame's *current* rule set (the state read in the basis that rule set fixes), so a superposition is rule-set-relative — definite in one rule set, superposed in another — as relational quantum mechanics (Rovelli, 1996) already holds. Because no rule set is privileged (the no-privilege principle again), the probability rule must be the *same* across every rule set, which is exactly the context-independence Gleason's theorem assumes. The framework thus supplies Gleason's premises — a vector-valued state (the linearity step above) and a context-independent measure (no privileged rule set) — from its own principles, and Gleason (Gleason, 1957) supplies the uniqueness: $|\cdot|^2$ is the only such measure in dimension ≥ 3, leaving the vector-structure rigour of the linearity step as the one posit.

The Hilbert posit, reconsidered: linearity as projection through frames. That one posit — the state lives in a complex Hilbert space and free evolution is unitary — is what treating Hilbert space *as a frame* is meant to reach. Read the inner product $\langle a \mid b\rangle$ as the projection of one frame's Narrative into another's rule set: how much of b shows up when b is read in the rule set a fixes. So the inner product is not imported structure but the framework's own

cross-frame reading. Linearity is then that projection distributing over components: during a free (empty-rule-set) segment no rule couples the Narrative's parts, so reading a superposition is superposing the readings — the same uncoupling, applied to the projection. Unitarity is that a free continuation is a *no-write*: it creates and destroys no distinction, so it preserves the Narrative and every cross-frame projection of it, norms and inner products conserved (the free-continuation conservation of Section 12). The Narratives, directional magnitudes that superpose, form the complex vector space; the inner product, a cross-frame map, makes it Hilbert; and its *completeness* — the defining Hilbert property — is the continuum living in the maps between frames, Hom(frame, frame), never in a frame's own stepped states (Section 2).

This meets the informational reconstructions of quantum theory, which *derive* the complex Hilbert space and the Born rule from a few operational axioms (Hardy, 2001; Chiribella et al., 2011). The framework appears to supply those axioms from its own principles: purification — every mixed state is pure in a larger system — is entanglement as a superframe dimension (Section 7); continuous reversibility is free continuation (a no-write is reversible); context-independence is no privileged rule set; and composition with local tomography is subframe nesting. If that mapping holds, Hilbert space and the Born rule follow by the cited reconstructions, as $|\cdot|^2$ followed from Gleason.

Honest boundary: this is the same move again, one level deeper, and the mapping has now been worked axiom by axiom against the Hardy and CDP reconstructions, with a result worth stating. The two *most* quantum axioms are supplied natively — Hardy's continuous reversible transformation by free continuation (a no-write is reversible, and the cycle-phase is continuous), and CDP's purification by entanglement as a superframe dimension (Section 7) — while most of the rest match native commitments (probability from realisation, compression and simplicity from the substrate's minimum description length, subspaces from subframe-as-frame). A closure-phase frame is likewise consistent with complex amplitudes at one system (the Bloch sphere, $K = 4$, rather than the real disk's $K = 3$), though a phase-less frame is equally constructible and the substrate does not require either. The remaining question — *local tomography*, which selects complex over real — is, correctly framed, *not a gap the substrate must close by forcing.* Almost everything in a graph substrate is optional: the compulsory core is structure, rules, subgraph-hood, and an admissibility grammar (the minimal-universe argument of the mind companion), and everything else — self-similarity, complex-over-real, the number of spatial dimensions, the gauge group — is contingent rule-content. The substrate therefore *admits* complex quantum theory (the self-similar complex composition gives $K_{AB} = 16$) rather than forcing it, exactly as it admits, without forcing, one value of N or one gauge group; a real-amplitude composition ($K_{AB} = 10$) is another admissible configuration, and our universe's selection of the complex one is self-location. Given that contingent choice the Born rule and Hilbert structure follow, which is how this paper treats all physics — shapes from contingent rule-content, the content never forced. The honest residual is then only technical. The operational-probabilistic instantiation — convex states, affine effects, a transformation monoid, and composition — is argued in the companion note on the Born recovery, from the substrate's weighted, ordered realisations together with the Narrative-momentum paper's first law (a frame carries its present structure forward until a write, so statistics depend only on the current state); it rests on the classical total-probability backbone rather than on any quantum assumption. Pending are the generalized-probabilistic-theory hygiene conditions: finite tomographic dimension, topological closure of the state space, and the full composition axioms. Nothing here requires that complex be derived: seeking to force complex over real would contradict the framework's own stance. Complex quantum theory has here the same status as $N = 3$ — admitted and self-located, not compelled. In a word, the framework *recovers* the Born rule rather than deriving it per se: it reproduces the known rule within its own structures — some steps forced given the premises (the squared modulus), the premises themselves admitted and contingent (complex, linear, self-similar) — not forcing it from necessity. Recovery, not derivation, is the honest verb and the right one: a contingent law

is recovered given its rule-content, never derived as compelled.

The reconciliation: Born, SR, and GR on one substrate

The recovery has a consequence larger than the Born rule. Special and general relativity are already reconciled here as one change-budget spent two ways (Proposition 4.1, Corollary 4.2), with c's invariance a theorem rather than a postulate (Proposition 2.1) — a theorem given the substrate's own units. Quantum collapse and relativity, the two historically unreconciled descriptions, are on this substrate not rivals but the in-frame *event* and the cross-frame *map* of one change-substrate, joined at c (Section 5). With the Born rule now *recovered* on the very same primitives — closure for the phase, no-privilege for phase- and context-independence, free continuation for the reversible structure — Born, SR, and GR stand reconciled on one substrate: the same c, the same no-privilege, the same closure loop underwrite all three.

Honest boundary: the strength is uneven — the SR–GR half a theorem given c-as-unit, the quantum–relativity half structural. This is emphatically *not* a quantum theory of gravity: no field is quantised, no metric is derived, no dimensionless constant is fixed. What is reconciled is the *conflict* — that quantum theory and relativity are two aspects of one change-substrate rather than incompatible frameworks — not the quantitative unification the scope explicitly disclaims.

9 Interference and the double slit

The double slit is obvious once two established facts are held together: causality is ordering *in a frame*, and a superframe applies changes to a subframe without passing through the subframe's admissibility.

Proposition 9.1 (Superposition is the possibility structure; interference is amplitudes combining by phase)**.** *Superposition is Graphic Equalisation's possibility structure $\mathcal{P}(G)$ — the admissible continuations a graph holds before collapse — so a particle "in superposition" is a graph whose $\mathcal{P}$ has more than one admissible continuation, none yet realised. Each continuation carries a directional Narrative (Section 8), an amplitude with a cyclic phase, and before collapse the alternatives' amplitudes combine, constructive where the phases align and destructive where opposed. The outcome weight is the combined closed loop, $|\sum \mathit{amplitude}|^2$.*

Two established facts make this un-mysterious. First, causality is ordering *in a frame*: there is no global fact about "which path the particle really took," because which-path is *undetermined*, not *undisclosed*, until a coupling realises it — a question with no answer of any kind until a coupling settles it. The "both paths" is not a particle in two places but the frame's $\mathcal{P}$. Second, a superframe applies changes to a subframe *without* passing through the subframe's admissibility (Graphic Equalisation: admissibility is self-governance, not armour — a graph's grammar governs its self-originated transformation, not world-caused forcing). So the interference is the superframe applying its combined-amplitude structure to the particle; the particle does not select a path by its own rules, and cannot veto the combination.

Corollary 9.2 (The double slit, duality, and delayed choice)**.** *Both paths are admissible continuations in $\mathcal{P}$; the superframe applies their combined amplitudes at the screen, giving fringes. A which-path coupling collapses before the screen — one continuation realised, the other no longer in the sum — so the superframe applies a single-path change and the fringes vanish. Wave–particle duality is the map/event distinction: the "wave" is the amplitudes combining (the map, the held $\mathcal{P}$), the "particle" the realised outcome (the event, one continuation collapsed), and "which is it really" names no difference. Delayed choice adds no retrocausality: collapse attaches to the actual interaction, not a retroactive rewrite (cost is a local write; no write reaches the past), so interference-or-not depends only on whether the which-path coupling occurs.*

Honest boundary: this grounds the shapes — superposition as possibility, interference as phase-combination, the double slit, duality, and delayed choice without retrocausality. The quantitative fringe pattern needs the amplitude phases computed, i.e. the Hilbert formalism, which is the Born-rule frontier of Section 8.

10 Dimensional constants are frame-units; dimensionless ones are not

The reading of c as a frame's own base unit generalises. Each dimensional constant is the base unit of one of a frame's own economies — 1 in-frame, a number only cross-frame.

- c is the change-rate unit (Section 2).
- $\hbar$ is the *aperture quantum*, the smallest resolvable cell: observation is funded by finite internal-change throughput, so fine resolution in one dimension trades against another and the trade floors at a minimum cell — which is the uncertainty principle, $\Delta x\,\Delta p \geq \hbar/2$ being the finite-aperture budget stated as a bound, with $\hbar$ its quantum.
- k_{B} is the information–energy conversion: Landauer's $k_{\mathrm{B}}T\ln 2$ (Landauer, 1961) — the floor cost of change of Graphic Equalisation's economics — is one bit's change converted from the information frame into the energy frame at that rate.
- G is the change-load–geometry coupling, mass deforming the containing frame.

That physics sets $c = \hbar = k_{\mathrm{B}} = G = 1$ (natural units) is then not a convenience but a recognition: each is a frame-unit, read as 1 in the frame's own terms with its familiar magnitude a cross-frame artefact (Proposition 2.1).

Proposition 10.1 (The dimensional/dimensionless boundary)**.** *This dissolves the* dimensional *constants and touches the* dimensionless *ones not at all. The fine-structure constant* $\alpha \approx 1/137$ *and the mass ratios cannot be set to* 1 *by any unit choice; they are genuine cross-frame* ratios*, real numbers with no dimension to absorb, and the framework says nothing about their values.*

It sharpens the real question rather than answering it: the numbers that need explaining are the dimensionless ones, while the pseudo-mysteries — why c, why $\hbar$ — are unit conventions. This is the standard observation that the number of fundamental *dimensional* constants is a matter of convention (Duff, 2002), landing on the side that they are units; the dimensionless remainder is left exactly as open as it was found.

11 The Planck scale, quantization, and fine-tuning

Three consequences follow. First, the *Planck scale* is where the dimensional units meet at 1 simultaneously ($\ell_P = \sqrt{\hbar G/c^3}$ and its companions): the aperture cell ($\hbar$), the change-rate ceiling (c), and the change-load–geometry coupling (G) reach their unit at once. Below it a frame cannot resolve, cannot order, and horizons its own geometry, so nothing there is representable — the " ? " of Graphic Equalisation's epistemic boundary with a location. The Planck scale *is* the frontier, which is why physics predicts its own breakdown there: not a failure of theories but the exhaustion of a frame's units, below which the pre-relational substrate begins.

Second, *quantization* is the aperture being cellular: because $\hbar$ is a minimum cell rather than an infinitesimal, action and energy come in $\hbar$-steps, so the "quantum" of quantum mechanics is Graphic Equalisation's discrete sufficiency read at the resolution axis — the granularity of the aperture, not a separate granularity of the world.

Third, the *dimensionless* constants are contingent rule-content: the framework cannot derive α or the mass ratios, but it can say what kind of thing they are — the contingent specifics of this substrate's rule-set (Graphic Equalisation's nothing-compulsory principle), so "why α" is not "why this necessary value" but "why this rule-set," answered by self-location (Proposition 14.2). This relocates fine-tuning at its root, which Section 14 takes up.

12 Symmetry and conservation

A symmetry is a transformation that produces no residual: a change the structure does not feel, free and reversible, a no-op in the container's grammar. Noether's theorem, symmetry $\leftrightarrow$ conserved quantity, is then the cost side of that. A quantity is conserved precisely when altering it would cost — produce residual, require a paid write — so the symmetric direction is free and the quantity perpendicular to it cannot move without paying. What a transformation leaves untouched is what it costs to change.

Proposition 12.1 (Momentum conservation is the free-continuation default)**.** *Space-translation symmetry — the rules the same everywhere — produces no residual under translation, and momentum is conserved. But momentum conservation is already the free-continuation default of Graphic Equalisation's momentum: the un-rewritten Narrative, Newton's first law, continuation costing nothing. Noether's momentum case and inertia are one statement — momentum is conserved because only changing it costs.*

Proposition 12.2 (Energy conservation is contingent on time-translation symmetry)**.** *Time-translation symmetry — the rules the same each cycle — conserves the change-budget, which is energy. The failure case is checkable: where the rules effectively change across cycles there is no time-translation symmetry and energy is not globally conserved. This is the standard general-relativistic result that a time-dependent spacetime has no global energy conservation — in an expanding universe photons redshift their energy away — here from the same principle: rules changing across cycles (cosmic expansion, or minting) means no conserved energy.*

Charge follows from gauge symmetry, which is a redundancy of description: a freedom in the representation that changes no physics, in the framework a cross-frame re-description (in the maps, where the continuous lives) producing no in-frame residual, so charge is conserved.

Remark (The gauge group is contingent rule-content)*.* The framework grounds *that* a symmetry yields a conservation law — the shape, residual-free $\leftrightarrow$ cost-to-change — but *which* symmetries hold, the $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ of the Standard Model, is a fact about this substrate's rule-set, not a necessity. "Why this gauge group" is "why this rule-set," the same relocation as "why α" (Section 11), answered by self-location (Proposition 14.2). The conservation *structure* is universal; the particular symmetries are this universe's.

13 The second law

Entropy and its arrow follow from the same economy.

Definition 13.1 (Entropy is what a frame does not distinguish)**.** *The entropy a frame holds is the log-count of the microstructure it carries but cannot tell apart — what falls below its aperture. Entropy is therefore frame- and aperture-relative from the start: a finer aperture resolves more and reads less entropy, a coarser one more.*

This is the Gibbs–Jaynes reading, in which entropy is a property of the observer's coarse-graining rather than of the world in itself (Jaynes, 1957); here it is grounded rather than assumed — entropy is *what this frame does not distinguish.*

Proposition 13.2 (The second law is no-copy accumulated)**.** *Every* write *discards its predecessor — the prior state goes* NULL *unless paid to keep — so each write pushes structure below the aperture, into the un-distinguished. (A free continuation is a no-write and discards nothing; it is the reversible, entropy-neutral case, which is why the arrow tracks writes and not mere succession.) Change therefore accumulates entropy, because change forgets by default and forgetting is exactly this demotion. The arrow of entropy is the no-copy asymmetry, which is the same asymmetry as accumulated change-cost (the Landauer floors summing) and as causation (a change enables its successor, not its predecessor). Equivalently, entropy is accumulated un-absorbed residual: what no grammar mints stays un-distinguished, so residual piling up is entropy rising.*

Corollary 13.3 (The demon is exorcised, and local order is affordable)**.** *Landauer's* $k_{\mathrm{B}}T\ln 2$ *(Section 10) is the price of* resolving *one bit — dragging one distinction up out of the un-distinguished, or erasing one back down. Maxwell's demon is exorcised by exactly this: it must pay to distinguish, and that payment is the entropy it thought it removed for free (Bennett, 1982). There is no free distinction, hence no free entropy reduction. Yet local entropy reduction is possible: minting resolves residual and builds structure, but it is a write, so it is paid, and the payment is Landauer heat exported to the surroundings — local order bought by global entropy increase. This is Schrödinger's "life feeds on negentropy" (Schrödinger, 1944) grounded: an organism, a learner, a growing graph is a region that pays to mint local structure and dumps the cost as heat. Life does not violate the second law; it is the second law with a subsidy.*

Remark (The arrows unify)*.* The thermodynamic arrow (entropy up), the causal arrow (change enables forward), and the psychological arrow (memory is present structure of a discarded past) are not three arrows that happen to align; they are one, the no-copy accumulation of change. Every "why do they agree" puzzle in the arrow-of-time literature is asking why three descriptions of a single asymmetry point the same way, and they do because the asymmetry is one: change does not keep its predecessor.

14 Cosmology: the initial state, the seed, and self-location

By Graphic Equalisation's no-start-end result the event order has no intrinsic first, so the beginning of the universe is not an intrinsic edge but a *chosen entry* — a boundary condition imposed from the containing frame. This is why physics supplies initial conditions separately from its laws: the laws are the ordering, which fixes no start, so the entry must come from outside. The Big Bang is that entry, the seed; what is "before" it is not an earlier time but the frontier, the pre-relational substrate, unrepresentable from within, the same " ? " that sits below the Planck scale (Section 11).

Proposition 14.1 (The Past Hypothesis is a seed, hence a hypothesis)**.** *The low-entropy initial state from which the second law takes its direction — the Past Hypothesis — is a seed: the initial rule-and-state configuration installed from the container. Its long-standing status as an assumption rather than a derivation is therefore correct in kind, since a seed cannot be derived from within (a mint cannot authorise its own installation). The arrow of time then splits cleanly: its* mechanism *is no-copy accumulation (Section 13), and its* origin *— that there is a low-entropy state to rise from — is the seed. The direction points away from the imposed low-entropy start; the running is no-copy.*

Proposition 14.2 (Why this seed: self-location)**.** *Why* this *seed — and, equally, why this* α*, this gauge group — is dissolved rather than declined, and what dissolves it is self-location. A map is referenced from "you are here," and the seed is the here: the origin the frame is drawn from, as* $c = 1$ *is the unit it is measured from. The observer is not independent of it — by frame-relative identity a different seed is a different observer — so "why this seed rather than another" assumes a trans-seed observer who could compare, and there is none (Axiom* -1*). It is*

that way because we are this way, and we are this way because it is that way: one determination read from inside, not two facts one explaining the other.

This inverts the fine-tuning inference. The constants are not tuned *to* a pre-existing observer; the observer is what those constants produce, so the apparent coincidence is the co-determination of observer and seed mistaken for a tuning aimed at a target. It is the same move as being surprised that the Graph Law applies to everything: it applies because that *is* what everything, measured, is — a constitutive identity, not a contingent breadth, the discriminator being independent re-derivation (Graphic Equalisation's composition falsifier). What remains out of reach is only *derivation*: no formula computes α or the seed. The *why* is answered; the *value* is not derived. Reference points take no "why" — they are where the why is asked.

15 Dimensions are contingent; time is optional

Time is not a must. A frame whose rules are not firing — inert, a rock — has no time at all: nothing happens in it, there is no succession, and in its own frame it merely *is*. What an external clock calls "later" is, for the rock, only another *is*; the weathering it undergoes is the *supergraph's* succession applied to it from above (admissibility being self-governance, not armour), not the rock's own. Time belongs to active, inferring frames — those with a firing closure loop to iterate (Graphic Equalisation's cycle-unit) and the inference to represent the succession. The rock has neither, so it has no time; its rules exist but are inert.

Proposition 15.1 (The observed time is one-dimensional)**.** *Every time we have observed is one-dimensional, and self-change gives a reason it tends that way: change is self-change on one structure, strictly ordered — no two of a frame's own changes are concurrent — so a frame lays down one succession at a time. But this is a tendency, not a theorem. Independent successions are no more prohibited than four spatial dimensions are; what selects the single observed time axis is the same filter that selects three spatial dimensions — decoherence and stability — not a necessity. So one-dimensional time is observed and selected, not forced; multi-dimensional time is unobserved, not forbidden. Time itself remains optional, absent from every inert frame.*

Space is N-dimensional by contingency: the number of positional coordinates the rules address is rule-content, so "why N" is "why this rule-set," answered by self-location (Section 14). Why $N = 3$ specifically is that self-location filtered by stability — Ehrenfest's result that stable orbits and stable atoms exist only in three spatial dimensions (Ehrenfest, 1917), so observers occur in the three-dimensional rule-sets and we are the observers such a rule-set produces. And the $3 + 1$ *signature*, time set apart from space, is the succession/position distinction: time is the change-axis, the one succession, space the config-axes, the many positions; one time and N space is one-succession-and-many-positions read as a signature.

The contingent-not-necessary status of these verdicts — and of the gauge group, the dimensionless constants, the arrow, and the value of c — is licensed by the minimal-universe argument (Section 1): a rock is a universe, and the minimal universe does without them, so none can be compulsory.

16 Identical particles are functional identity

Graphic Equalisation's Axiom -1 holds that two things with no difference that makes a difference are the same, and its no-frame-free-identity theorem that one cannot say which of two is which independently of an individuating frame. This is the quantum treatment of identical particles exactly: two electrons carry no label, and "which is which" names no difference and so has no content. Classical physics labels identical objects — treats two identical bodies as distinct in principle — and that label is a fiction, a distinction imposed where none exists.

Proposition 16.1 (Quantum statistics, and the Gibbs paradox, are functional identity)**.** *Classical Maxwell–Boltzmann counting labels identical particles and over-counts arrangements; quantum Bose–Einstein and Fermi–Dirac counting treats them as one where they are indistinguishable. The Gibbs paradox — the spurious entropy of mixing two identical gases — resolves by not labelling them, which is Axiom* -1*, and ties to Section 13: entropy is what a frame distinguishes, so identical particles, having nothing to distinguish, add no mixing entropy. The* $1/N!$ *Gibbs correction is functional identity made quantitative.*

Corollary 16.2 (Pauli exclusion is "distinct things need a distinction")**.** *If two fermions were in the same state — the same in every quantum number — they would carry no distinguishing distinction, and by Axiom* -1 *would be one, not two. To be two they must differ somewhere. The exclusion principle is therefore the requirement that distinct things be distinguishable: there is no such thing as two indistinguishables in one state, because that "two" names no difference. Exclusion is the arithmetic of identity, not a force.*

Remark (The honest limit: the identity argument does not select fermions)*.* The corollary's argument is weaker than it looks, and the boson case shows why: it applies verbatim to two bosons in one state, which is observed and which the Gibbs count above needs to be genuinely two. So Axiom -1 does not *derive* exclusion — exclusion is *consistent* with it, and what actually distinguishes the fermion case from the boson case (the minus sign under exchange, tied to a 2π rotation) is the spin–statistics theorem, which needs structure this account does not supply. Nor does "same in every quantum number" by itself individuate: two electrons in distant atoms differ by position, and it is the full state including that which the argument must range over. The framework grounds *that* particles are indistinguishable, and hands the fermion/boson split back untouched.

17 Positioning and prior art

The pieces are individually established; the contribution is the conjunction under one change-economy and the mechanisation of shapes prior art postulates, geometrises, or offers heuristically.

Discrete substrates and emergent spacetime. Several programmes derive relativistic structure from a discrete substrate, and two are closer to the change-budget of Section 3 than any continuum account. D'Ariano and Tosini (2013) obtain the metric from *pure event-counting* on topologically homogeneous causal networks, with time dilation appearing as an increased density of foliation leaves within one tick of a clock and length contraction as the corresponding decrease in event density. That is the same move made here and made earlier, and the counting reading of dilation is theirs, not this paper's. Three things are added rather than repeated. Their result requires *topological homogeneity* of the network; here no homogeneity is assumed, because a cycle's length varying with the rules and the graph is the mechanism rather than an irregularity to be assumed away. Their counting delivers the metric; the budget additionally *splits* — throughput spent on motion or on internal change — which is what yields mass as reserved budget, general relativity as change-load rather than only the special-relativistic metric, and the equivalence principle as the seam between the two (Section 4). And their result is a result about spacetime; the same economy is run here through collapse, entanglement and the amplitude weights (Sections 5 to 8), and belongs to a substrate of which physics is one embodiment rather than to physics alone. Knuth and Bahreyni (2014) derive the Minkowski metric and the Lorentz transformations from consistent quantification of a causally ordered set with respect to an *embedded observer*, explicitly without assuming the constancy of c; that is the closest existing result to the reading of c given above, and it reaches it by a quantification argument rather than by a throughput economy. The Wolfram model (Wolfram, 2020; Gorard, 2020a) derives general covariance from causal invariance and Lorentz covariance from flat foliations, but stipulates

c-constancy in its definition of edge lengths (Section 2) and, in the quantum companion (Gorard, 2020b), introduces the Born weights as an a-priori rule rather than recovering them — which is the respect in which Section 8 differs from it. Causal set theory (Bombelli et al., 1987; Surya, 2019) takes order and counting as primitive without rewriting dynamics. The honest summary is that emergence of relativistic structure from a discrete substrate is established prior art with several independent routes, and this paper does not claim the emergence. What it claims is narrower and, so far as these sources go, unoccupied: one budget, split two ways, carrying the special and general cases together and continuing into the quantum content of the same substrate without a further postulate at either seam.

Deriving c, and mass as internal motion. That c should be *derived* rather than postulated has precedent by a different route. Ignatowsky (Ignatowsky, 1911) and, cleanly, Lévy-Leblond (Lévy-Leblond, 1976) obtain the Lorentz transformation from the relativity principle together with homogeneity, isotropy, and group structure alone, with an invariant speed *emerging* rather than assumed and its value left empirical; those derivations use symmetry as the engine where the present one uses the change-substrate — c as each frame's own change-throughput unit, 1 by construction. The reading of mass as the budget reserved for internal change has two standing precedents, offered heuristically rather than mechanised: Schrödinger's *zitterbewegung*, the electron's rest energy as internal motion at c (Schrödinger, 1930), developed as a rest-mass-from-internal- circulation interpretation by Hestenes (Hestenes, 1990); and the elementary result that confined massless radiation — light in a box — carries rest mass E/c^2, internalised c-motion appearing as inertia. The budget-split itself is the constant-magnitude four-velocity of Minkowski geometry (Minkowski, 1909).

Relativity as postulate. Einstein's special (Einstein, 1905) and general (Einstein, 1916) relativity take c-invariance and the equivalence principle as postulates; the present account renders the first a theorem (Proposition 2.1) and the second a common cause (Section 4), at the cost of claiming only the shapes.

Entropy and the demon. The observer-relative reading of entropy is Jaynes's (Jaynes, 1957); the resolution of Maxwell's demon by the cost of information is Landauer's and Bennett's (Landauer, 1961; Bennett, 1982); life as locally paid negentropy is Schrödinger's (Schrödinger, 1944). The present account ties all three to one substrate law — cost is the write.

Dimensional constants. That the count of fundamental *dimensional* constants is conventional is Duff's position in the Duff–Okun–Veneziano exchange (Duff, 2002); the present account gives the reason (they are frame-units) and draws the boundary at the dimensionless remainder.

18 Falsifiability commitments

Each is exposed to attack; the scope caveat of Section 1 applies throughout.

1. c **as frame-unit.** That a frame reads its own base rate as 1 is definitional and not itself at risk; the falsifiable claim is the cross-frame step. Fails if two frames in relative motion can be exhibited reading *different* c for the same signal — that is, if aperture does not self-calibrate — or if c-invariance requires a postulate that cannot be recovered as the unit reading.

2. **One budget, two dilations.** Fails if velocity time dilation and gravitational time dilation cannot be exhibited as the same fixed budget spent on motion and on change-load respectively — if the equivalence principle is not their common cause here.

3. **Dimensional/dimensionless boundary.** Fails if a genuinely dimensionless constant can be set to 1 by unit choice, or if a dimensional constant carries an irreducible value not absorbable into a frame's units.

4. **Second law as no-copy.** Fails if entropy can rise in a frame whose *writes* preserve their predecessors, or fall in one without a paid write exporting cost. (Free continuations are no-writes and so are the entropy-neutral case; the claim concerns writes, which are the only priced, predecessor-discarding operations.)

5. **Indistinguishability is functional identity.** Fails if identical particles carry a frame-free label distinguishing them, or if the Gibbs $1/N!$ correction and Fermi exclusion cannot be read as "distinct things require a distinguishing distinction." (The boson/spin–statistics content is not claimed.)

6. **Measurement is frame-relative collapse.** Fails if a frame-independent fact about whether collapse has occurred is required to describe a Wigner's-friend scenario — if the friend's outcome and the outside frame's superposition cannot both stand as the event and the map of two frames. (The Born-rule probabilities are not claimed.)

7. **Entanglement is a superframe dimension.** Fails if the correlation of an entangled pair is a subframe-local quantity rather than a joint dimension NULL in each subframe, or if a reduced single-particle state can be pure — i.e. if the mixedness is not the inaccessibility of a superframe dimension. (The Tsirelson bound and correlation strengths are not claimed.)

8. **Conservation is the cost side of symmetry.** Fails if a continuous symmetry (a residual-free transformation) carries no conserved quantity, or if a conserved quantity persists across a genuine change in the rules — in particular, if global energy is conserved in a time-dependent (rules-changing) regime such as cosmic expansion. (The gauge group is claimed to be contingent, not derived.)

9. **The initial state is a seed, answered by self-location.** Fails if "why this seed / constant / gauge group" can be posed from a trans-frame standpoint that could have obtained a different one — if there is an observer-independent "we" separable from the seed. (The claim is that the *why* is self-location and the *value* is not derived; both together, not either alone.)

10. **Amplitude is directional Narrative; Born reduces to linearity.** Fails if the complex amplitude cannot be identified with a magnitude-times-direction operative-history structure, or if the Born square admits no closed-loop (conjugate) reading. Sharper: given complex, *linearly superposing* amplitudes and no privileged phase, $|a|^2$ is forced (by phase-independence plus the additivity of decohered alternatives), so the claim fails if all of those premises hold yet the probability is not the squared modulus. It fails also if the mapping onto the informational reconstructions cannot be made good. (Linearity is *not* claimed derived; the residue is stated in Section 8.)

11. **Interference is possibility plus superframe forcing.** Fails if interference requires the particle's own admissibility to select among paths, or if a which-path coupling that collapses before the screen does not remove the fringes — i.e. if superposition is not the possibility structure and the superframe does not apply the combination without passing through the subframe's grammar.

12. **Time is optional; the observed time is one-dimensional.** Fails if an inert frame (no firing rules) nonetheless has an in-frame succession. (That the observed time is 1-D is claimed decoherence-selected, not forced — multi-dimensional time is unobserved, not prohibited; that space is N-D with $N = 3$ is likewise contingent and self-located. The minimal-universe

licence, the block universe, free will, personal identity, and death are treated in the companion mind paper and carry their falsifiers there.)

Scope note (not a falsifier). The paper claims shapes only; it would be overclaiming if any passage were read to derive a field equation, a dimensionless constant, or a quantum-gravity result — none is asserted.

References

John S. Bell. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1:195–200, 1964.

Charles H. Bennett. The thermodynamics of computation — a review. *International Journal of Theoretical Physics*, 21(12):905–940, 1982.

Giulio Chiribella, Giacomo Mauro D'Ariano, and Paolo Perinotti. Informational derivation of quantum theory. *Physical Review A*, 84(1):012311, 2011.

Giacomo Mauro D'Ariano and Alessandro Tosini. Emergence of space–time from topologically homogeneous causal networks. *Studies in History and Philosophy of Modern Physics*, 44(3):294–299, 2013.

Michael J. Duff. Comment on time-variation of fundamental constants. arXiv:hep-th/0208093, 2002. (Duff–Okun–Veneziano, "Trialogue on the number of fundamental constants.")

Paul Ehrenfest. In what way does it become manifest in the fundamental laws of physics that space has three dimensions? *Proceedings of the Amsterdam Academy*, 20:200–209, 1917.

Albert Einstein. Zur Elektrodynamik bewegter Körper. *Annalen der Physik*, 322(10):891–921, 1905.

Albert Einstein. Die Grundlage der allgemeinen Relativitätstheorie. *Annalen der Physik*, 354(7):769–822, 1916.

Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6:885–893, 1957.

Jonathan Gorard. Some relativistic and gravitational properties of the Wolfram model. *Complex Systems*, 29(2):599–654, 2020.

Jonathan Gorard. Some quantum mechanical properties of the Wolfram model. *Complex Systems*, 29(2):537–598, 2020.

Lucien Hardy. Quantum theory from five reasonable axioms. arXiv:quant-ph/0101012, 2001.

David Hestenes. The zitterbewegung interpretation of quantum mechanics. *Foundations of Physics*, 20(10):1213–1232, 1990.

Wladimir A. von Ignatowsky. Einige allgemeine Bemerkungen zum Relativitätsprinzip. *Physikalische Zeitschrift*, 11:972–976, 1911.

Kevin H. Knuth and Newshaw Bahreyni. A potential foundation for emergent space-time. *Journal of Mathematical Physics*, 55(11):112501, 2014.

Edwin T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106:620–630, 1957.

Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17:59–87, 1967.

Luca Bombelli, Joohan Lee, David Meyer, and Rafael D. Sorkin. Space-time as a causal set. *Physical Review Letters*, 59(5):521–524, 1987.

Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.

Jean-Marc Lévy-Leblond. One more derivation of the Lorentz transformation. *American Journal of Physics*, 44(3):271–277, 1976.

Hermann Minkowski. Raum und Zeit. *Physikalische Zeitschrift*, 10:75–88, 1909. Address delivered 1908.

Sumati Surya. The causal set approach to quantum gravity. *Living Reviews in Relativity*, 22:5, 2019.

Stephen Wolfram. A class of models with the potential to represent fundamental physics. *Complex Systems*, 29(2):107–536, 2020.

Carlo Rovelli. Relational quantum mechanics. *International Journal of Theoretical Physics*, 35:1637–1678, 1996.

Erwin Schrödinger. Über die kräftefreie Bewegung in der relativistischen Quantenmechanik. *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse*, 24:418–428, 1930.

Erwin Schrödinger. *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, Cambridge, 1944.